SUMMARY

- The current, I flowing in a conductor $I = \frac{dQ}{dt}$, where dQ is the charge that flows through a cross-section in a time interval dt. SI unit of current is ampere (A). 1A = 1 C s^{-1}.
- The current density J in a conductor is the current flowing per unit area. $\left(J = \frac{I}{A}\right)$
- Current is a scalar but current density is a vector.
- The general form of Ohm's law $\vec{J} = \sigma\vec{E}$
- Practical form of Ohm's law states that V $\propto$ I, or V = IR where I is the current.
- The resistance R of a conductor is $R = \frac{V}{I}$. SI unit of resistance is ohm (Ω) and $1\ \Omega = \frac{1V}{1A}$
- The resistance of a material $R = \rho\frac{l}{A}$ where l is length of the material and A is the area of cross section.
- The resistivity of a material determines how much resistance it offers to the flow of current.
- The equivalent resistance (R_S) of several resistances (R_1, R_2, R_3……..) connected in series combination is R_S = (R_1+R_2 +R_3……..)
- The equivalent resistance (R_P) of several resistances (R_1, R_2, R_3……..) connected in parallel combination is $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ...$
- Kirchoff's first rule (Current rule or junction rule): The algebraic sum of the currents at any junction is zero.
- Kirchoff's second rule (Voltage rule or loop rule): In a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit.
- Electric power is the rate at which energy is transformed.
- If a current I flows across a potential difference V, the power delivered to the circuit is P = IV.
- In a resistor R, the electrical power converted to heat is P = $I^2R = \frac{V^2}{R}$
- The energy equivalent of one kilowatt-hour (kWh) is 1kWh = 3.6 X 10^6 J.
- Metre bridge is one form of Wheatstone's bridge.
- Potentiometer is used to compare potential differences.
- Joule's law of heating is H = VIt (or) H = I^2Rt.
- Thermoelectric effect: Conversion of temperature differences into electrical voltage and vice versa.

UNIT-2(XII-Physics_Vol-1).indd 119 04-03-2019 11:01:58